

Table 4. Regime-conditioned within-family selection used after Definition 4.1 chooses a watermark family. For each estimated edit regime, the concrete scheme is the method with the highest regime-conditioned score $S_r(m)$.

Family	Estimated edit regime	Regime-conditioned score $S_r(m)$ with tuple (AUC, z)	Selected representative	Watermarking Parameters
Token-level	$\hat{\epsilon} \approx 0.25$	HCW: 1.137 (0.910, 3.40) DiPMark: 1.104 (0.900, 3.90) HeavyWater: 1.072 (0.880, 4.20) SimplexWater: 1.052 (0.870, 4.50) Unigram: 0.982 (0.880, 8.80) KGW: 0.954 (0.860, 9.60)	HCW	δ -reweight variant (method=delta)
Semantic	$\hat{\epsilon} \approx 0.42$ $\hat{\epsilon}_s \approx 0.10$	PMark: 1.305 (0.850, 1.20) SemStamp: 1.220 (0.820, 1.50) SimMark: 1.170 (0.800, 1.70)	PMark	Rejection parameter $\rho = 0.25$
Distribution-preserving	$\hat{\epsilon} \approx 0$	CGW: 1.990 (0.990, -5.80)	CGW	Security parameter $\lambda = 128$, fixed secret key across runs

Note. This table explains the second stage of the Hybrid. Definition 4.1 first selects the family from the estimated edit regime. The concrete scheme is then chosen within that family using the regime-conditioned score $S_r(m) = \text{AUC}_r(m) + \frac{1}{1 + \max(z_r(m), 0)}$. The token-level row uses the DIPPER regime at $\hat{\epsilon} \approx 0.25$. The semantic row uses the back-translation regime, for which $(\hat{\epsilon}, \hat{\epsilon}_s) \approx (0.42, 0.10)$. Greener cells indicate higher within-family scores and therefore the selected representative at that regime.